

FYP Proposal

AI-Based Real-Time Classroom Noise Classifier & Smart Auto Notes Generator

1. Project Title

AI-Powered Real-Time Classroom Noise Detection and Automatic Lecture Notes Generator

2. Introduction

Classroom environments often suffer from background noise, overlapping speech, and disturbances that reduce the quality of learning and make it difficult for students to understand or revisit lectures. Many students rely on recorded lectures, but background noise and unclear audio reduce the usefulness of these recordings.

This project proposes an AI-powered classroom assistant that can:

1. Classify noise types in real time.
2. Filter the noise intelligently.
3. Transcribe the teacher's speech.
4. Automatically generate summarized lecture notes.

3. Problem Statement

Traditional lecture recordings often contain background noise, poor audio quality, missing summaries, and no structured notes. Students waste time re-listening to long recordings. Teachers lack tools to provide clean recordings or automatic notes.

4. Objectives

Primary Objectives:

- Detect and classify classroom noise in real time.
- Separate teacher's voice from background noise.
- Correct audio quality automatically.
- Transcribe teacher speech.
- Generate summarized lecture notes.

Secondary Objectives:

- Provide GUI for students and teachers.
- Save audio, transcripts, and notes for future use.

5. Scope of the Project

Included: audio classification, noise filtering, transcription, summarization, UI.

Not Included: multi-language support, hardware device.

6. Significance of the Study

Helps students get clean notes, helps teachers save time, and supports universities in digital learning.

7. Methodology

1. Data Collection
2. Noise Classification Model (CNN/CRNN/Audio Transformers)
3. Real-Time Noise Filtering (DSP + AI)
4. Speech Recognition (Whisper)
5. Notes Generation (LLM Summarization)
6. System Integration (Python backend + UI)

8. Tools & Technologies

Python, PyTorch, Whisper, Transformers, Librosa, Streamlit/FastAPI.

9. Expected Output

Noise classification, clean audio, transcript, auto-generated notes.

10. Feasibility

Models run on normal laptops; datasets publicly available.

11. Timeline

Requirements – 1 week

Dataset – 2 weeks

Model – 3 weeks

ASR – 2 weeks

Summarizer – 2 weeks

Integration – 3 weeks

Testing – 2 weeks

Report – 2 weeks

12. Conclusion

A unique AI solution combining audio AI, ML, and NLP to improve classroom learning.